

LESSON 08 / 第八課

The Urban Forest as a Population: From Canopy Cover to the Stem 作為族群的城市森林：從樹冠覆蓋到單株

Format. Short literature review. Claims that carry a number or a causal ranking are tied to a numbered source. Scale runs from the largest published pattern to the finest measured process. The lesson is not sequenced from the previous number and is not a regional case file unless the object itself is regional.

體例。短文獻綜述。帶數字或因果排序的陳述繫到編號文獻。尺度由最大已發表格局寫到最細已測過程。本課不由上一號遞推；除非對象本身是地域性的，否則不是單一地區檔案。

Abstract

摘要

An urban forest is a population living in a built energy and water regime, not a collection of isolated specimens that happen to share a city. At city scale the published metrics are canopy cover and its distribution. The 3-30-300 statement is an access heuristic, not a biome law. At neighbourhood scale, equity of cover and of species mix become visible. At the stem, infrastructure conflicts are usually upstream failures of selection and space, and a storm is a stress test of the population already planted. A planting target without a survival budget (Lesson 23) overstates future canopy. Carbon and cooling claims inherit the same demography: dead trees do not transpire.

城市森林是生活在建成能量與水體制中的族群，不是碰巧共享一座城的孤立標本集合。城市尺度上，已發表度量是樹冠覆蓋及其分布。3-30-300 陳述是可及性啟發式，不是生物群系定律。街坊尺度上，覆蓋公平與樹種組合變得可見。單株上，基礎設施衝突通常是選擇與空間的上游失敗，而風暴是對已種下族群的壓力測試。沒有存活預算的種植目標（第二十三課）會高估未來樹冠。碳與冷卻主張繼承同一人口學：死樹不蒸騰。

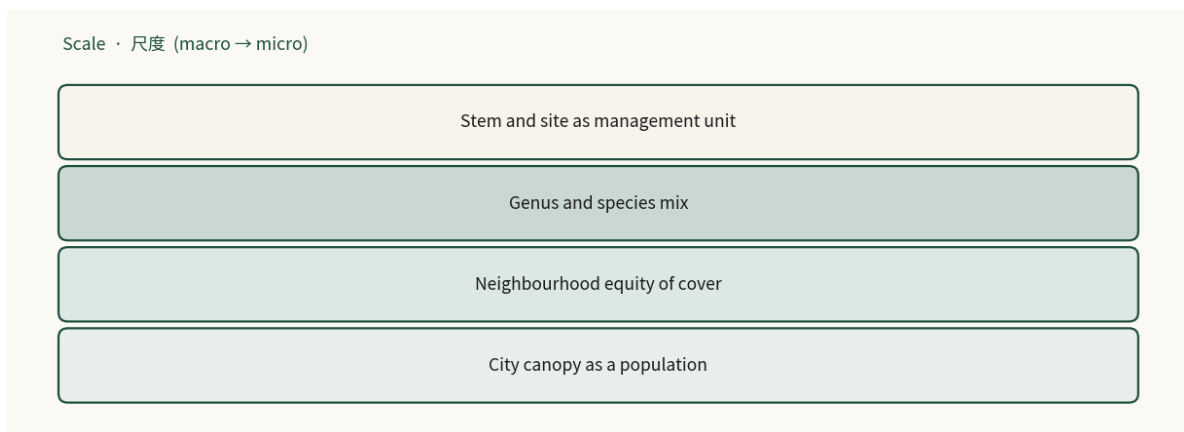


Figure 1. Nested scales from city canopy to the management stem.

圖 1. 由城市樹冠到管護單株的嵌套尺度。



Figure 2. Cover, access and diversity are different numbers.

圖2. 覆蓋、可及與多樣性是不同數字。

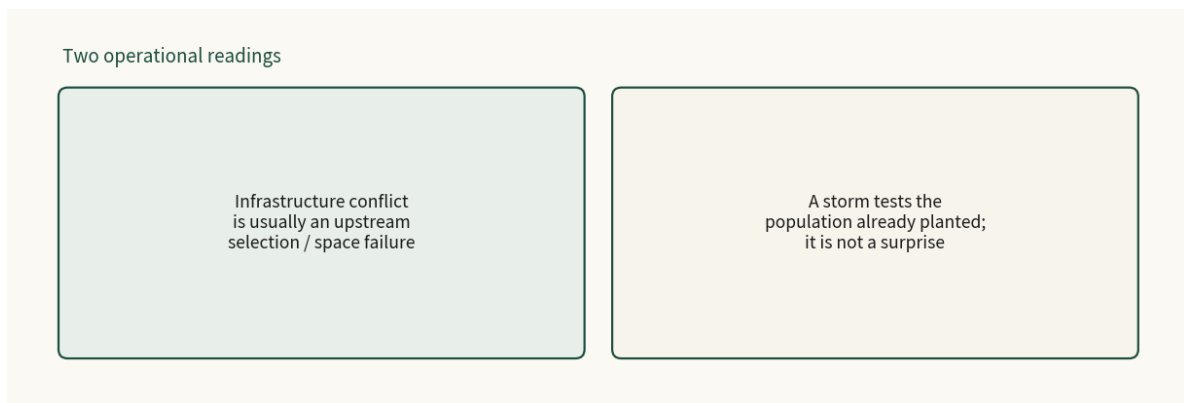


Figure 3. Conflicts and storms read the population already planted.

圖3. 衝突與風暴讀出已種下的族群。

1. Scope

1. 範圍

The object is the urban forest as a published population system: cover, equity, diversity, and the stem as the unit on which those numbers are spent.

對象是作為已發表族群系統的城市森林：覆蓋、公平、多樣性，以及這些數字所花在其上的單株。

2. Cover, access, diversity

2. 覆蓋、可及、多樣性

Canopy-cover estimates depend on the sensor, the height threshold and the date [1]. They are not interchangeable with leaf-area index or with cooling. Konijnendijk's 3-30-300 statement (3 trees from every home, 30% neighbourhood cover, 300 m to a park) is an access heuristic assembled from public-health and forestry literatures; it is not a measured universal optimum [2]. Santamour's 10-20-30 ceiling remains the usual diversity heuristic [3]. Cities whose inventories collapse to a short list fail it as a population before a specialist pest arrives [4].

樹冠覆蓋估計取決於傳感器、高度閾值與日期 [1]。它們不可與葉面積指數或冷卻互換。Konijnendijk 的 3-30-300 陳述（每家看見 3 棵樹、街坊 30% 覆蓋、300 公尺到公園）是從公共衛生與林業文獻組裝的可及性啟發式；不是測得的普遍最適 [2]。Santamour 的 10-20-30 上限仍

是常見多樣性啟發式 [3]。清查塌縮到短名單的城市，在專性有害生物到來之前就已作為族群未通過 [4]。

3. Equity, infrastructure, storms

3. 公平、基礎設施、風暴

Cover is uneven inside cities. Neighbourhood-scale work treats that unevenness as an equity object, not as an accident of taste [5]. Infrastructure conflict (heave, sewer, sightline) is usually the mature expression of an upstream selection or space failure, not a mysterious property of ‘trees’. Extreme wind is a stress test of damping, root plates and residual geometry already in the population. Planting numbers without a survival budget overstate the canopy those numbers claim [6].

覆蓋在城市內部不均。街坊尺度工作把該不均當成公平對象，而不是品味事故 [5]。基礎設施衝突（抬升、下水道、視距）通常是上游選擇或空間失敗的成熟表達，不是「樹」的神秘屬性。極端風是對族群中已有的阻尼、根盤與剩餘幾何的壓力測試。沒有存活預算的種植數字，會高估那些數字所主張的樹冠 [6]。

4. Limits

4. 限度

i-Tree and similar tools translate structure into service estimates under declared assumptions. The assumptions are part of the result. A carbon or cooling number that cannot name its mortality and irrigation regime is not yet a management number.

i-Tree 及類似工具在已宣布假設下把結構譯成服務估計。假設是結果的一部分。不能說出其死亡與灌溉體制的碳或冷卻數字，還不是管理數字。

Study questions

研習題

Choose the option that retains the published result. Answers follow the last item.

選出保留已發表結果的選項。答案在最後一題之後。

1. An urban forest is treated here as:

1. 此處把城市森林視為：

- A. Isolated specimens only.
- A. 只有孤立標本。
- B. A population in a built energy and water regime.
- B. 建成能量與水體制中的族群。
- C. A t/R table.
- C. t/R 表。
- D. A palm meristem.
- D. 棕櫚分生組織。

2. 3-30-300 is:

2. 3-30-300 是：

- A. A biome law.
- A. 生物群系定律。

B. An access heuristic.

B. 可及性啟發式。

C. A P50.

C. 一個 P50。

D. Wall 4.

D. 第 4 壁。

3. Canopy cover and cooling:

3. 樹冠覆蓋與冷卻：

A. Are interchangeable numbers.

A. 是可互換數字。

B. Are not interchangeable.

B. 不可互換。

C. Equal occupancy.

C. 等於佔用。

D. Equal flash density.

D. 等於閃擊密度。

4. Inventories that collapse to a short taxon list:

4. 塌縮到短分類群名單的清查：

A. Pass 10-20-30 automatically.

A. 自動通過 10-20-30。

B. Fail the diversity ceiling as a population.

B. 作為族群未通過多樣性上限。

C. Raise safety margins.

C. 提高安全邊際。

D. Replace targets.

D. 取代目標。

5. Infrastructure conflict is usually:

5. 基礎設施衝突通常是：

A. A mysterious property of trees.

A. 樹的神秘屬性。

B. An upstream selection or space failure.

B. 上游選擇或空間失敗。

C. Proof of zero canopy.

C. 零樹冠的證明。

D. A lightning path.

D. 雷電路徑。

6. A storm in this review is:

6. 本綜述中的風暴是：

A. An inexplicable surprise.

A. 無法解釋的意外。

B. A stress test of the population already planted.

B. 對已種下族群的壓力測試。

C. A mulch specification.

C. 覆蓋物規格。

D. Wall 1.

D. 第 1 壁。

7. Planting numbers without a survival budget:

7. 沒有存活預算的種植數字：

A. Forecast canopy exactly.

A. 精確預測樹冠。

B. Overstate future canopy.

B. 高估未來樹冠。

C. Equal P88.

C. 等於 P88。

D. Are QTRA.

D. 是 QTRA。

8. Dead trees:

8. 死樹：

A. Still transpire at catalogue rates.

A. 仍按目錄速率蒸騰。

B. Do not transpire.

B. 不蒸騰。

C. Raise mycorrhizae automatically.

C. 自動提高菌根。

D. Set SOS.

D. 設定 SOS。

9. i-Tree service numbers:

9. i-Tree 服務數字：

A. Are assumption-free.

A. 沒有假設。

B. Inherit declared assumptions, including demography.

B. 繼承已宣布假設，包括人口學。

C. Replace inspection.

C. 取代檢查。

D. Are t/R laws.

D. 是 t/R 定律。

10. Neighbourhood unevenness of cover is treated as:

10. 覆蓋的街坊不均被視為：

A. Pure taste.

A. 純品味。

B. An equity object.

B. 公平對象。

C. A cable grade.

C. 鋼纜等級。

D. A nursery system.

D. 苗圃系統。

References

參考文獻

- [1] Nowak D.J., Greenfield E.J. urban tree cover assessment papers.
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- [4] Raupp M.J. et al. street-tree diversity and pest vulnerability.
- [5] Nesbitt L. et al. urban green equity literature.
- [6] Hilbert D.R. et al. (2019). Urban tree mortality. AUF 45: 167–200.
- [7] McPherson E.G. et al. municipal forest structure and function.

Closing note. This lesson is written as a short review article. Disagreement in the literature is kept in view where the results actually diverge.

結語。本課以短綜述文章寫成。文獻真正分歧之處，兩造結果都保留在視野內。